Database Development and Design (SOE201)

Lecture 1a: Module Introduction

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AIAT

Learning Outcomes

- Overview of Module

Module Information

- Code and title: *SOE201* - Database Development and Design (~~CPT201~~)
- Length: One semester
- Credit value: 5
- Teaching delivery: 13 Lectures + 12 labs (practicals, tutorials and seminar)
- Teaching assistants
  - Mr. Zimu Wang and Ms. Nijia Han
- Office hour: 11:00-13:00, Mondays and 13:00-15:00, Thursdays
- Coursework date/time: on module handbook

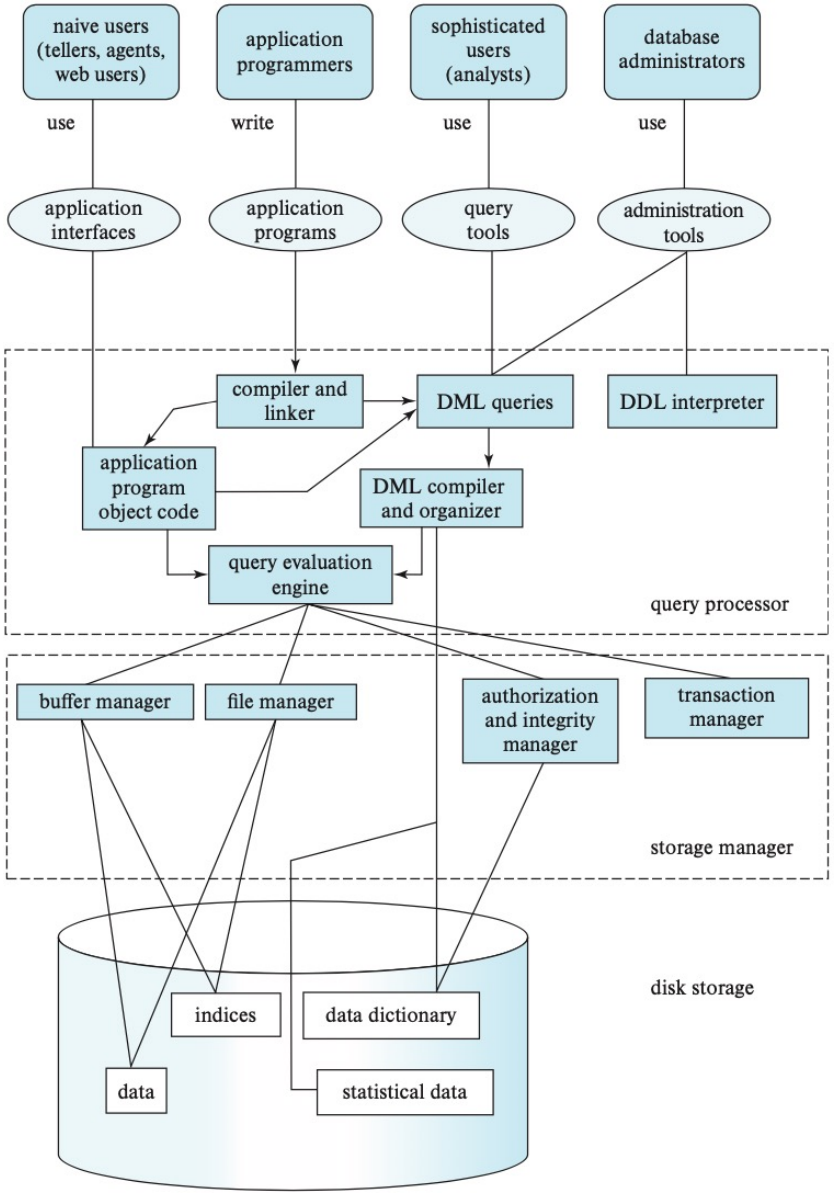
Assessment

- Assessment
  - Two coursework
    - Coursework 1 – in-class test (15%): indexing techniques
    - Coursework 2 - lab reports: (25%)
  - Final exam (60%)
    - Part A: short answer questions
      - 20 questions, 2 marks each, in total 40 marks
    - Part B: Problem-Solving and Quantitative Questions
      - 2-3 'big' questions, 30+30 or 30+20+10, in total 60 marks
- Resit exam (100%)

What Have Been Covered Before

- Database Development Lifecycle;
- Conceptual Design;
- Logical Design and Physical Design;
- Normalisation;
- SQL: DDL, DCL, DML;
- Relational Algebra;
- Different DBMS Environments;
- Legal Frameworks;
- etc…

RDBMS Components



Lecture Topics

- W1-Lecture1a - Introduction to Module
- W1-Lecture1b - Data Storage Structures
- W2-Lecture2a - Indexing Techniques
- W2-Lecture2b - B+Tree Indexing
- W3-Lecture3a - Hash Indexing
- W3-Lecture3b - Advanced Indexing
- W4-Lecture4a - Introduction to Relational Model
- W4-Lecture4b - Query Evaluation Basics
- W5-Lecture5a - Query Evaluation: Selection
- W5-Lecture5b - Query Evaluation: Join
- W6-Lecture6a - Query Optimisation 1
- W6-Lecture6b - Query Optimisation 2
- W7 – CONSOLIDATION WEEK
- W8-Lecture7a - Transaction Management 1
- W8-Lecture7b - Transaction Management 2
- W9-Lecture8a - Concurrency Control
- W9-Lecture8b - Failure Recovery
- W10-Lecture9a - Web Technologies and Data Storage 1
- W10-Lecture9b - Web Technologies and Data Storage 2
- W11-Lecture10a - Object-Oriented Database
- W11-Lecture10b - Distributed Database
- W12-Lecture11a - Big Data Storage (tentative guest lecture)
- W12-Lecture11b - Blockchain-based Storage (tentative guest lecture)
- W13-Lecture12 - Data Analytics
- W13-Lecture13 - Revision

Labs

- Practicals
  - 1st hour for task 1; 2nd hour for task 2 (*AI NOT allowed*)
  - Topics: JDBC and indexing, Java Transactions, XML processing, RDF for knowledge graph query, ontology and linked data, big data storage MongoDB.
- Tutorials
  - 1st hour, try to solve all questions by yourself or group; 2nd hour, discuss solutions
  - Content of tutorials aligned with lectures
  - Topics: storage, query evaluation, query optimisation, transaction management, feedback to coursework, prepare for exam
- Seminar
  - Database design

Learning Outcomes

- A. Critically analyse and compare advanced database techniques, such as indexing, query processing and optimisation;
- B. Explain and apply principles underpinning transaction management within relational DBMS;
- C. Effectively use and compare recent Web technologies and data representation formalisms, including both semi-structured and structured data, in developing database solutions;
- D. Distinguish and analyse principles underlying object-relational databases, distributed databases, big data storage systems and blockchain-based storage systems;
- E. Explain main concepts in data warehousing and data mining, and demonstrate their uses in real-world applications.

Recommended Textbooks

- Compulsory
  - Database System Concept, 7th Ed, A. Silbershatz, H.F. Korth and S. Sudarshen
- Optional
  - Database Systems: The Complete Book, Hector Garcia, Jeffrey D. Ullman, Jennifer Widom
  - Database Systems: A Practical Approach to Design, Implementation, and Management, T. Connolly and C. Begg

End of Lecture

- Reading
  - Module handbook
  - FAQs
  - Papers
    - Cameron R. Jones, Benjamin K. Bergen, "Large Language Models Pass the Turing Test", arXiv:2503.23674
    - Savannah Thais, "Misrepresented Technological Solutions in Imagined Futures: The Origins and Dangers of AI Hype in the Research Community", https://arxiv.org/pdf/2408.15244v1
    - Konstantine Arkoudas, "GPT-4 Can't Reason", https://arxiv.org/abs/2308.03762v2
    - Ziwei Ji et al, "Survey of hallucination in natural language generation", ACM Computing Surveys, 2023
    - Yue Zhang et al, "Siren's song in the AI ocean: A survey on hallucination in large language models", arXiv preprint arXiv:2309.01219, 2023
    - Parshin Shojaee et al, "The Illusion of Thinking: Understanding the Strengths and Limitations of Reasoning Models via the Lens of Problem Complexity," arXiv:2506.06941
    - Nataliya Kosmyna et al, "Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task", arXiv:2506.08872